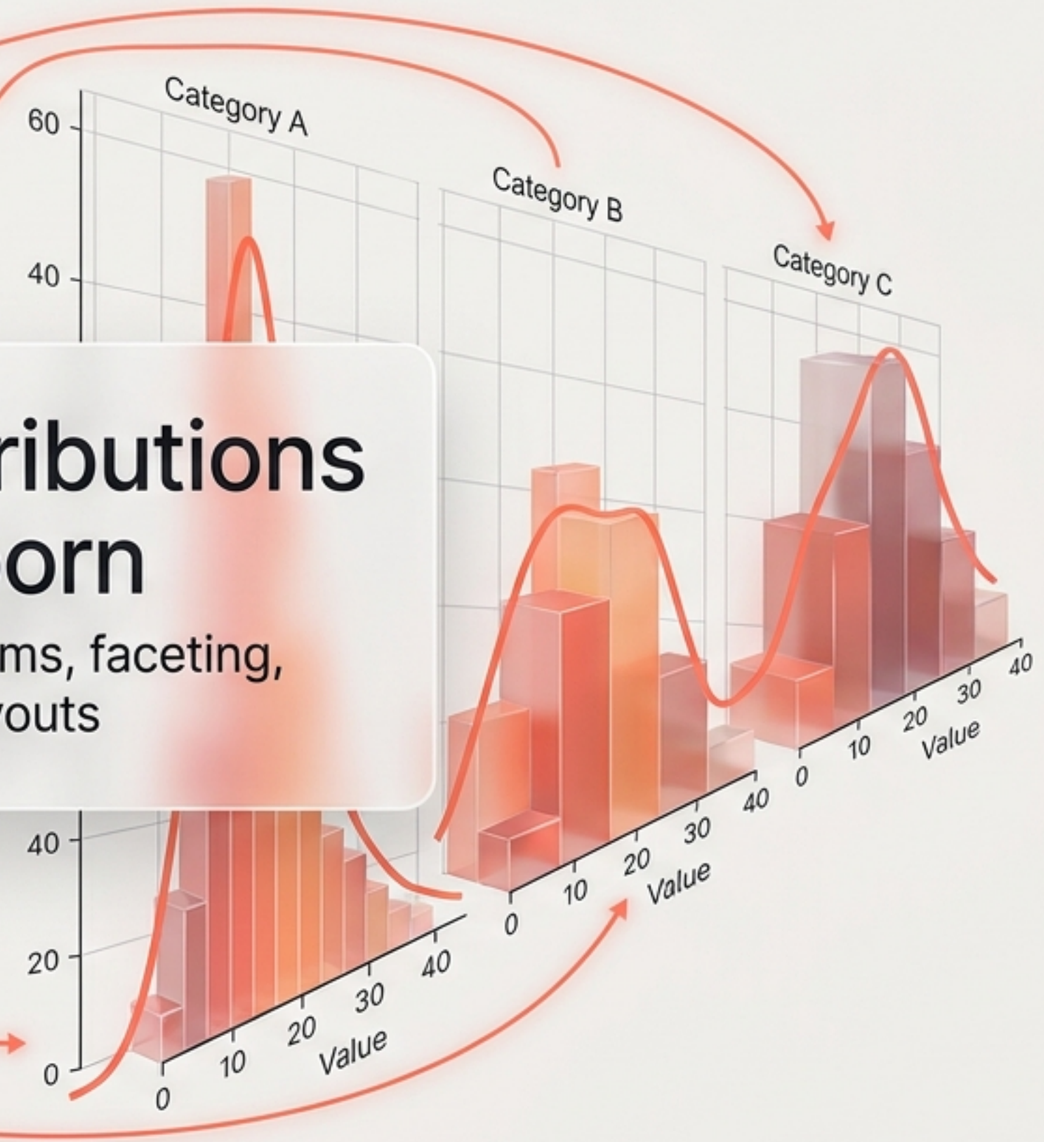
```
                (
{                       =   }
      [  ]   (  =   -     +  ÷
      +     [    ]    *  /      )
}
plt.figure()
sns.histplot(data,
            bins=30,
            kde=True,
            color='coral',
            hue=plot,
            )
plt.fgure()
g = sns.FacetGrid(
        col='category',
        height=5,
        aspect=1.5
        hue=5
    g.map(g, g)

plt.show()
                { =     ]   [
     +  _ )   - *  / +       (       ]
}    /          .       )
                }
```

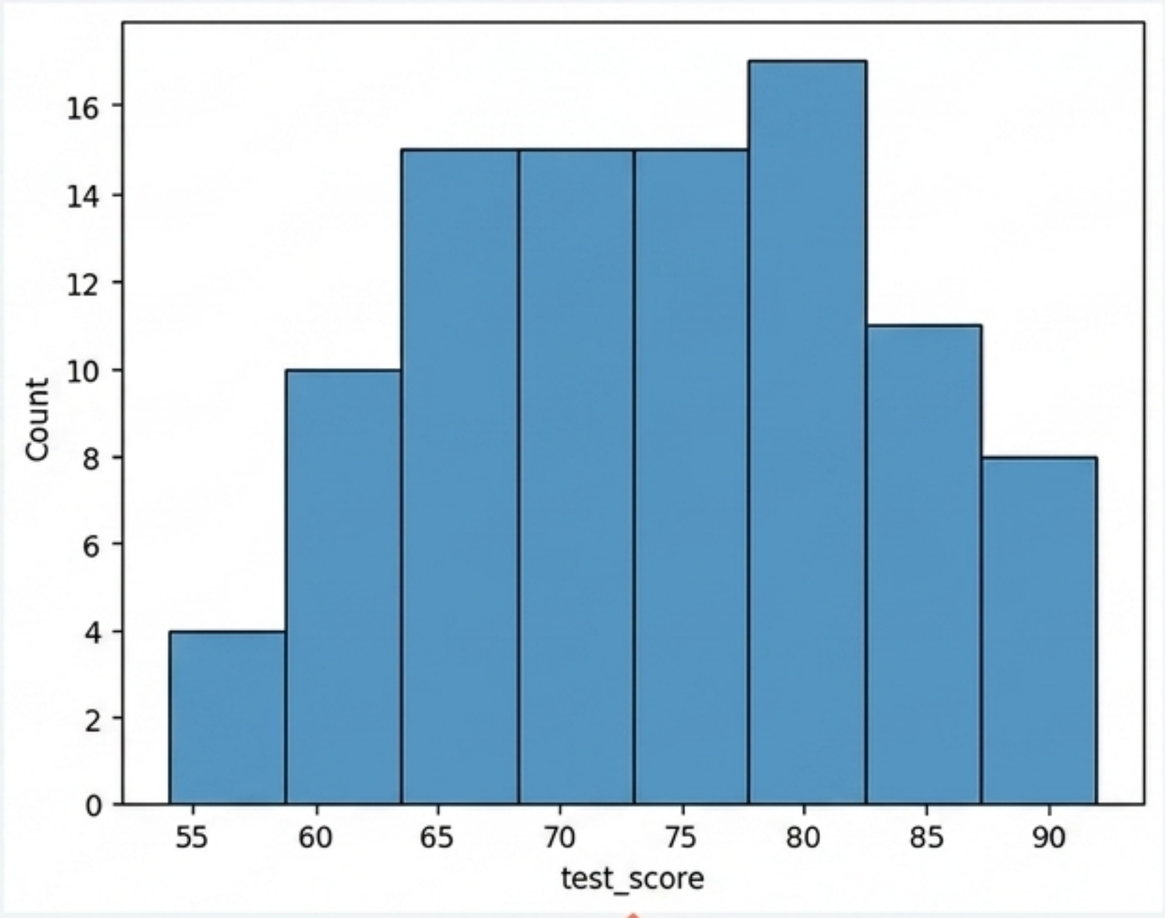
Mastering Distributions with Seaborn

A visual guide to histograms, faceting, and multi-plot layouts



01 / The Foundation

```
sns.histplot(data=student, x="test_score")
```

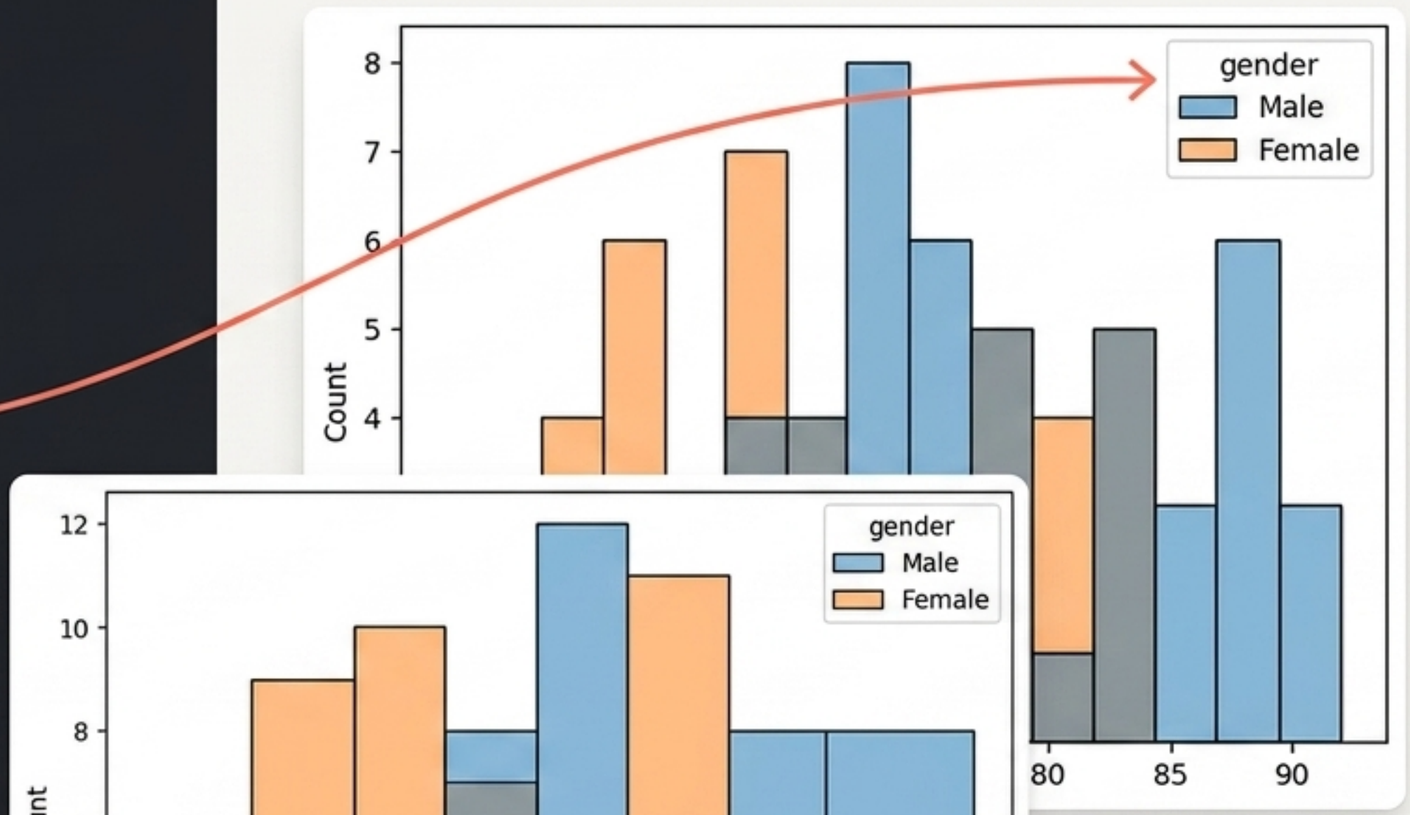


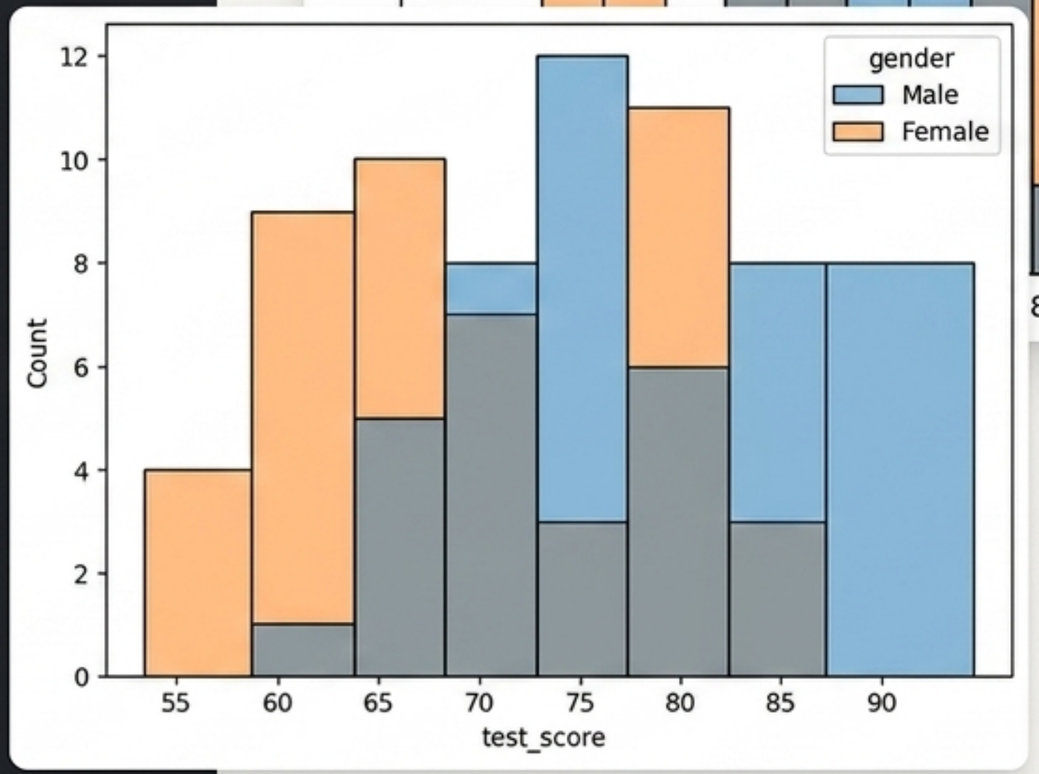
Axes-Level Function

`histplot` automatically calculates bin sizes and counts the frequency of the target variable. It is the fundamental building block for continuous data distribution.

02 / Stratification

```
sns.histplot(data=student,
             x="test_score",
        hue="gender")
```





Grouping with hue

The hue parameter splits the dataset into distinct subgroups within the same axes. Overlapping areas are handled by default using transparency and blending.

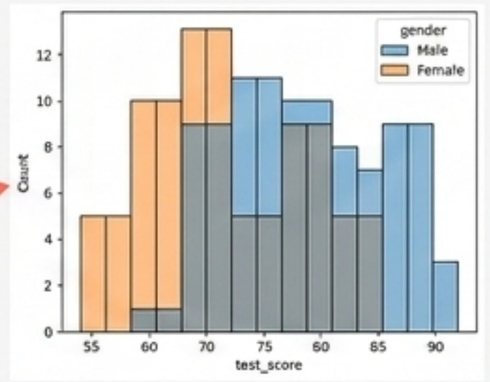
03 / Granularity

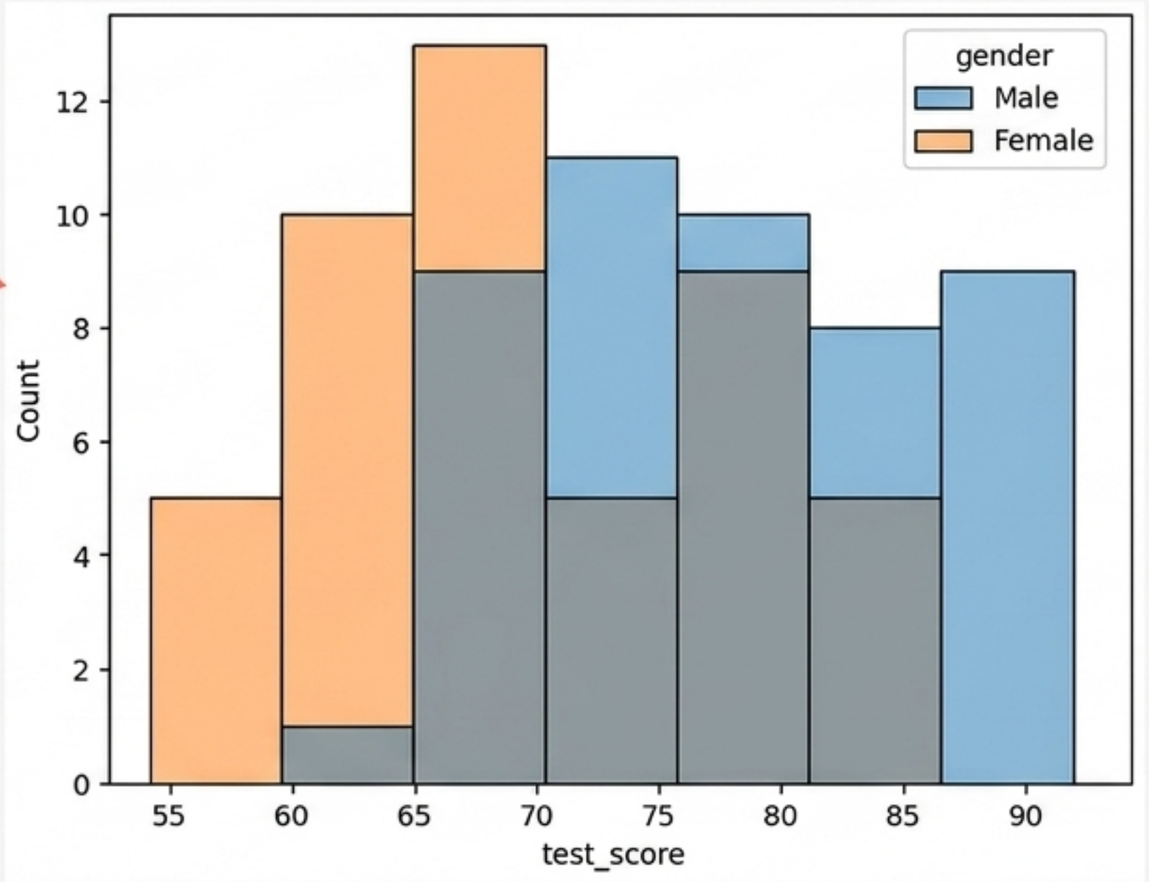
A

```
sns.histplot(..., hue="gender", bins=15)
```

B

```
sns.histplot(..., hue="gender", bins=7)
```





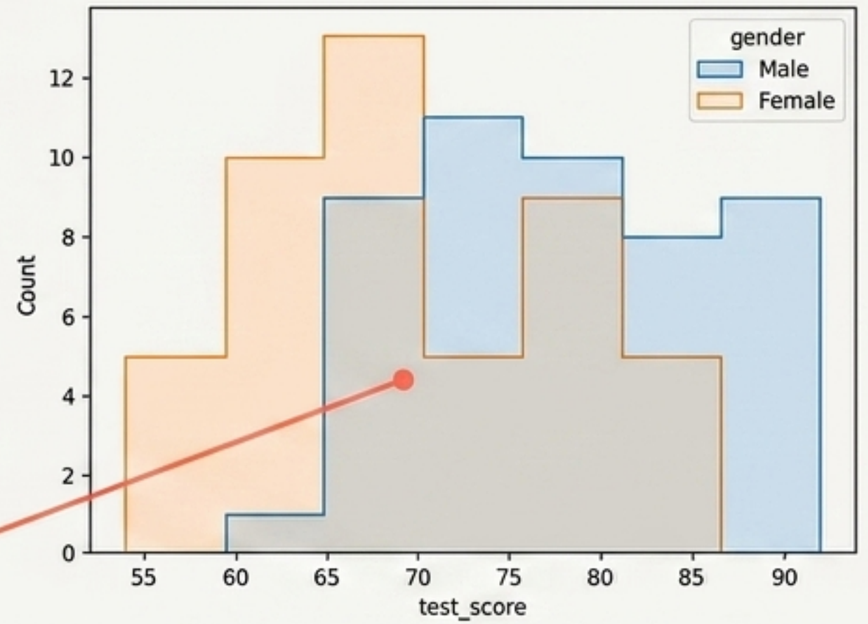
Controlling Bins

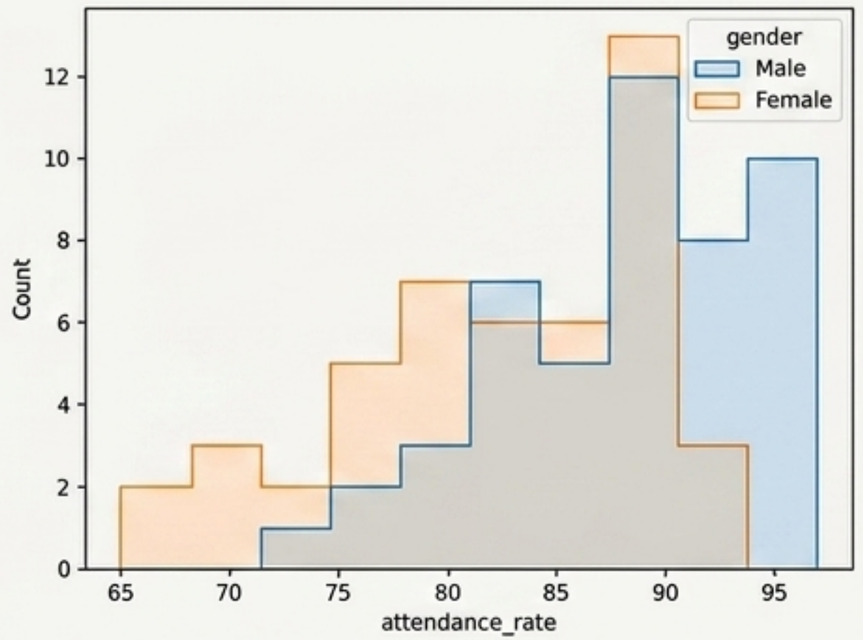
Lowering the bin count aggregates data into wider ranges. This smooths out noise but obscures fine detail. Higher bin counts reveal tighter, more specific clusters.

04 / Aesthetics

```
sns.histplot(...,
             hue="gender",
             bins=7,
             element='step')

sns.histplot(...,
             x="attendance_rate",
             hue="gender",
             bins=10,
             element='step')
```





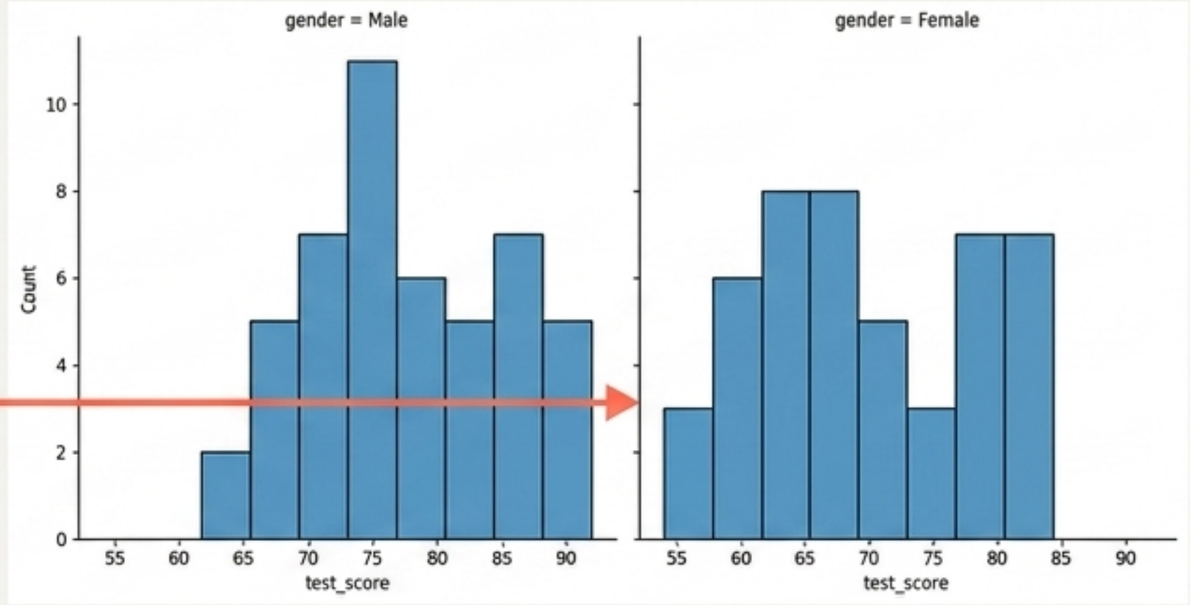
The Step Element

Changing the visual geometry from solid bars to outlines (step) removes internal dividers. This dramatically improves readability for overlapping distributions.

05 / Faceting

```
sns.displot(data=student, x="test_score",
            bins=10, col="gender")

sns.displot(data=student, x="study_hours",
            bins=10, col="gender")
```



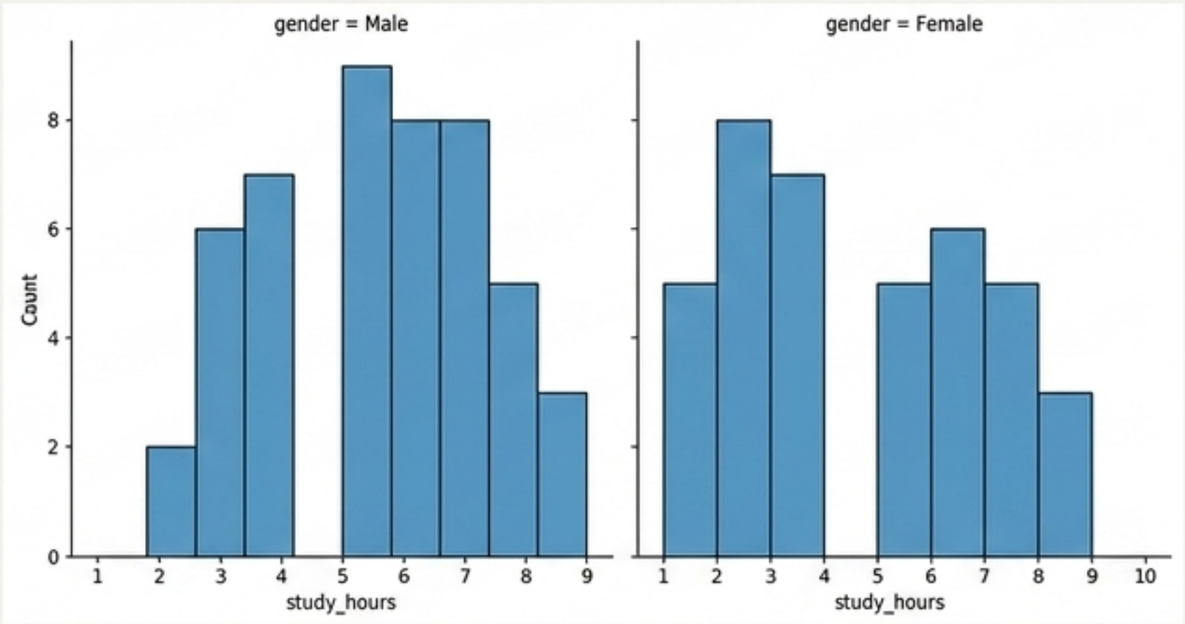
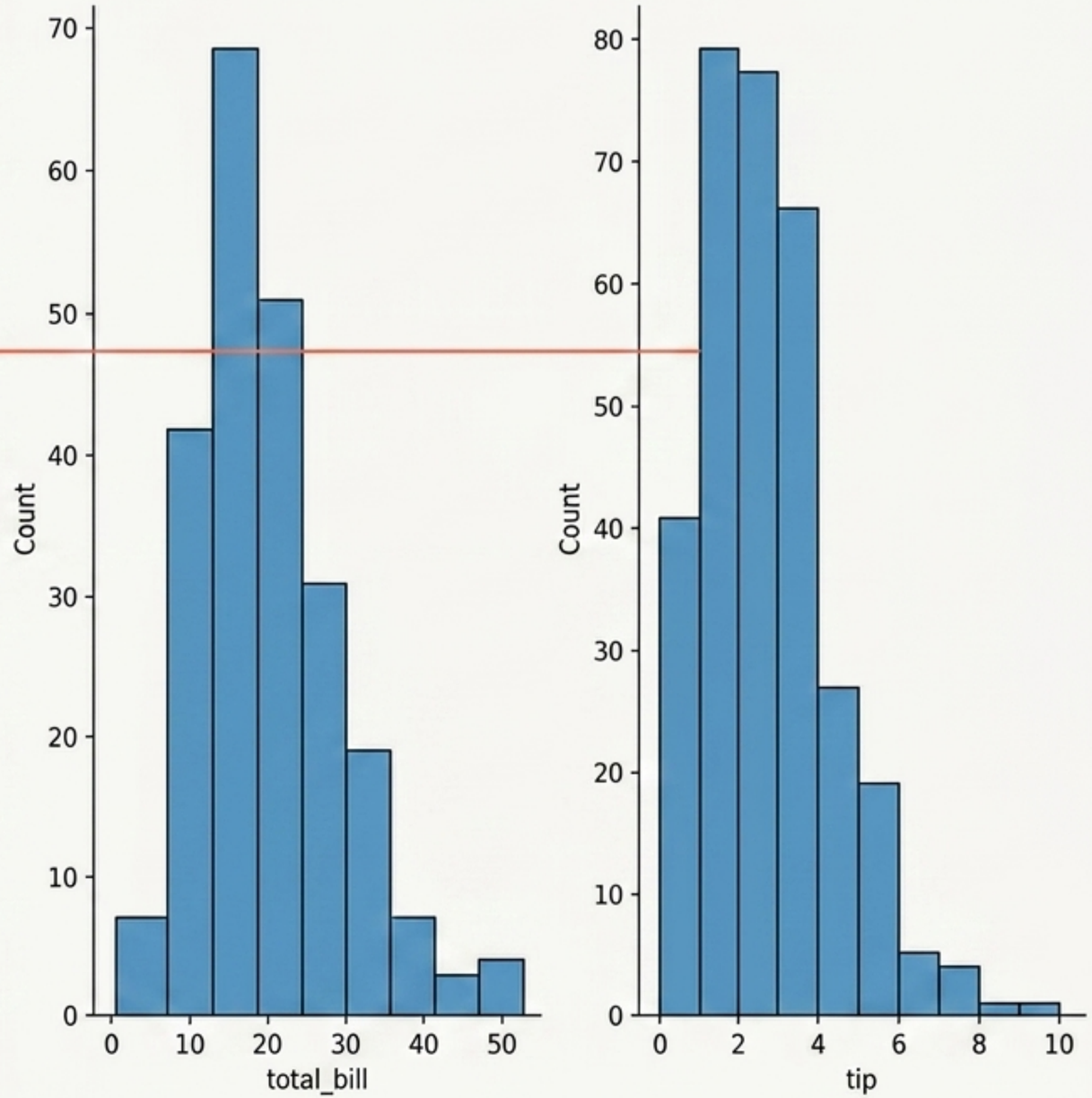


Figure-Level Faceting
`displot` is a **Figure**-level function. Instead of overlapping data on one axis, `col="gender"` generates instant "small multiples"—separate, synchronized subplots sharing exact axis limits for accurate side-by-side comparison.

06 / Applying to New Contexts

```
sns.displot(data=tips_data, x='total_bill', bins=10)
sns.displot(data=tips_data, x='tip', bins=10)
```

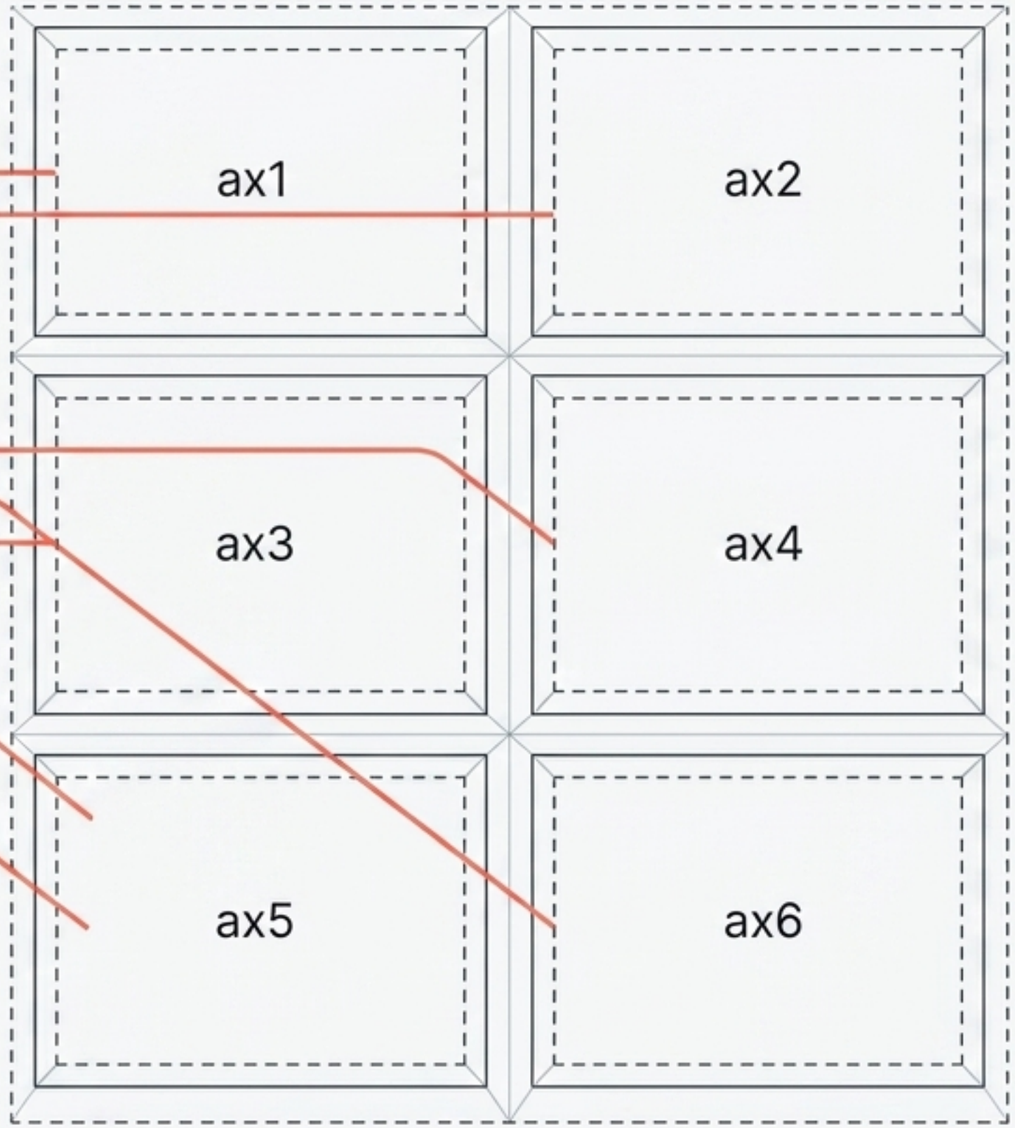


Validating the Technique

The exact same displot architecture instantly reveals the right-skewed behavior of restaurant spending and tipping in a completely different dataset structure.

07 / Grid Setup

```
fig, ((ax1, ax2), (ax3, ax4), (ax5, ax6)) =
    plt.subplots(3, 2, figsize=(12, 12))
```



The Multi-Plot Blueprint

Matplotlib's subplots(3, 2) dictates a grid of 3 rows and 2 columns. Tuple unpacking allows us to assign a specific variable name to each physical 'box' on the canvas.

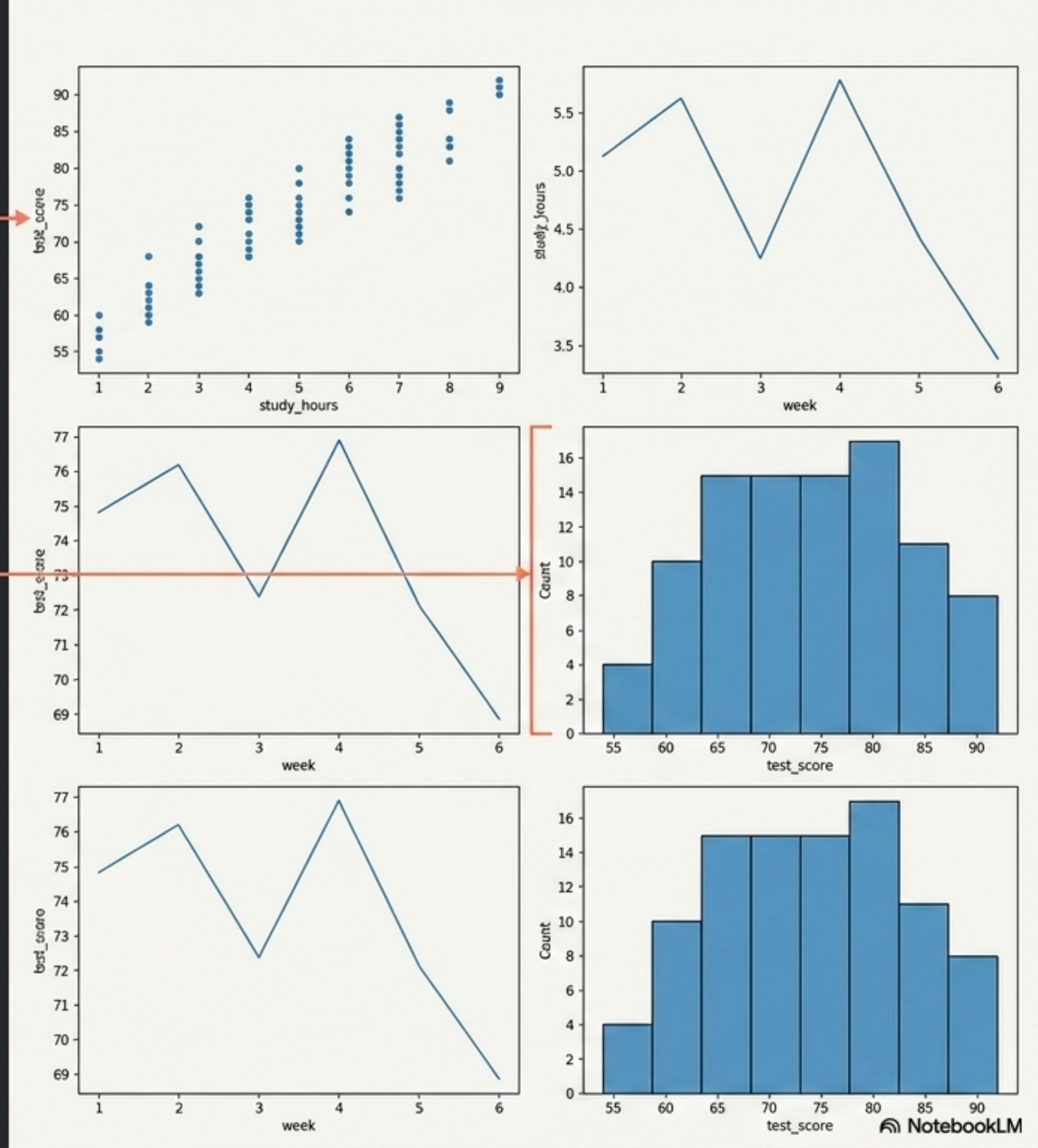
08 / Compositing

```
sns.scatterplot(..., ax=ax1})
sns.lineplot(..., ax=ax2})
sns.lineplot(..., ax=ax3})
sns.histplot(..., ax=ax4})
sns.histplot(..., ax=ax6})
sns.histplot(..., ax=ax6})
```

```
plt.tight_layout()
```

Populating the Dashboard

By using the ax= parameter, we can force individual axes-level plots into specific positions within our custom blueprint.



Synthesis: The Seaborn Matrix

Axes-Level Functions

Examples: `histplot()`, `scatterplot()`, `lineplot()`

Behavior: Draws onto a single, specific matplotlib axis.

Superpower: Precision compositing. Perfect for slotting into custom multi-plot dashboards via the `ax=` parameter.

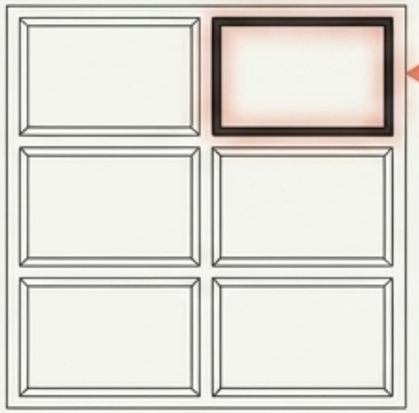


Figure-Level Functions

Examples: `displot()`, `relplot()`, `catplot()`

Behavior: Initializes its own complete Figure environment.

Superpower: Instant faceting. Exclusive access to `col=` or `row=` parameters to generate synchronized small multiples grids without manual wireframing.